

AYMANE AIT DADS

Research Intern - AI Agents & Efficiency | LLM Fine-Tuning · Agentic Systems · Reasoning Efficiency
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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT and staged adapter continuation training - directly demonstrating frontier AI efficiency research at scale. Diagnosed that hard-task failures in cipher and bit-manipulation reasoning stemmed from reasoning over-expansion rather than incorrect reasoning, a task-level error analysis insight with direct implications for agentic system reliability and token-efficiency. Brings hands-on experience in LLM fine-tuning (LoRA, PEFT, SFTTrainer, vLLM), agentic pipeline design (LangChain, tool-calling, RAG, MCP server), ablation methodology, and chain-of-thought dataset engineering aligned with M365 Research priorities. Aims to advance AI efficiency and agentic systems research at Microsoft by applying controlled fine-tuning techniques and diagnostic evaluation frameworks to large-scale, product-impact problems.

CORE COMPETENCIES

AI Efficiency Research | Agentic Systems | LLM Fine-Tuning | Chain-of-Thought Reasoning | Ablation Studies | Transformer Models | Research Prototyping | Model Evaluation

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered automated feature engineering and model evaluation pipeline on **100K+ fiber-optic network records**, reducing manual analysis time by approximately 60% and directly enabling the network optimization team.
- Built Random Forest classification and K-Means clustering models to map customer performance profiles across **5 commercial service tiers**, producing outputs consumed directly by operational stakeholders.
- Designed a clustering pipeline integrating upload speed, latency, and jitter signals to classify network quality across **thousands of nodes**, applying ensemble methods and unsupervised learning in tandem.
- Delivered stakeholder presentations translating model outputs into maintenance recommendations, bridging **research findings and product impact** for a non-technical audience at scale.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge (In Progress)

Kaggle Competition · 2025-2026 · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Performed **task-level error analysis** across unit conversion, cipher, symbolic, gravity, and bit manipulation categories; built a diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode, with direct implications for agentic AI efficiency.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM ImSecu + NTIRE 2026 CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M parameters) with a custom MLP head, dual learning rates, and FP16 mixed precision on 250K images spanning 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; discovered that 1-epoch training (0.793 AUC) outperforms 4-epoch training (0.767 AUC), demonstrating that early stopping improves out-of-distribution generalization for large pretrained models.

PyTorch · OpenCLIP · Hugging Face Transformers · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Statistics, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Reasoning: LoRA | QLoRA | SFTTrainer | PEFT | vLLM | chain-of-thought | Mamba-Transformer | completion-only loss | BF16/FP16

Agentic AI & ML Engineering: LangChain | RAG | tool-calling | agentic pipelines | MCP server | Claude API | prompt engineering | PyTorch | Hugging Face Transformers

Data Science & MLOps: Python | Pandas | Scikit-learn | ablation studies | Git | CUDA debugging | Triton | Docker | SQL